

Adaptive Fairness-Constrained Decision Making via Offline Reinforcement Learning

Progress Report
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Members

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1. Introduction

Machine learning systems are increasingly deployed in high-stakes decision-making domains such as criminal justice, lending, and hiring. While these systems improve efficiency and scalability, they are often susceptible to systematic biases across demographic groups, raising concerns about fairness and accountability. To address this, prior work has proposed fairness-aware techniques such as threshold adjustment, reweighting, and post-processing corrections. However, most of these approaches are inherently static, assuming a fixed underlying data distribution.

In practice, real-world environments are non-stationary. Demographic proportions and feature distributions evolve over time, leading to shifting fairness–accuracy trade-offs. As a result, fairness interventions that are optimal at one point may degrade under distribution shift, causing previously fair models to become biased. This limitation motivates the need for adaptive fairness strategies that can respond dynamically to changing conditions.

In this project, we formulate fairness mitigation as a sequential decision-making problem. Specifically, we model the problem as a Markov Decision Process (MDP), where the state consists of classifier performance and subgroup fairness metrics, and actions correspond to threshold adjustments applied to different subgroups. This formulation enables an agent to continuously adapt its decisions in response to observed fairness violations and performance changes.

To learn such adaptive policies, we adopt an offline reinforcement learning (RL) framework using Conservative Q-Learning (CQL). We construct a custom fairness environment with offline logged trajectories generated via scripted behavior policies, and simulate demographic distribution shifts to model non-stationarity. This setup enables stable policy learning without requiring online interaction.

We evaluate our approach on the COMPAS recidivism dataset, using AIF360 to compute standardized fairness metrics, including Equal Opportunity gap and Demographic Parity gap. Preliminary results indicate that the learned adaptive policy significantly reduces fairness disparities compared to static baselines, while maintaining competitive predictive performance.

These findings suggest that modeling fairness mitigation as a dynamic control problem can provide a more robust approach to balancing fairness and accuracy in non-stationary environments.

2. Problem Formulation

We model fairness intervention as a Markov Decision Process (MDP) defined by the tuple (S, A, R, T) , where an agent makes sequential decisions over a fixed-length episode of 50 timesteps.

2.1 State Space (S)

At each timestep t , the agent observes a 10-dimensional state vector capturing current classifier performance and subgroup fairness metrics:

$$s_t = [\text{accuracy}, \text{TPR_AA}, \text{TPR_Cauc}, \text{TPR_Hisp}, \text{EO_gap}, \text{DP_gap}, \text{prop_AA}, \text{prop_Cauc}, \text{prop_Hisp}, t/T]$$

where TPR denotes the True Positive Rate per group, EO_gap is the Equal Opportunity gap ($\max \text{TPR} - \min \text{TPR}$ across groups), DP_gap is the Demographic Parity gap, group proportions reflect the current demographic distribution under drift, and t/T is a normalized time index encoding drift progress.

2.2 Action Space (A)

The agent selects from 9 discrete threshold intervention actions as described in Table 1.

Table 1: Action Space Definition

Action	Description
0	No intervention
1	Increase African-American threshold
2	Decrease African-American threshold
3	Increase Caucasian threshold
4	Decrease Caucasian threshold
5	Increase Hispanic threshold
6	Decrease Hispanic threshold
7	Increase all thresholds globally
8	Decrease all thresholds globally

Each threshold adjustment applies a fixed step size of 0.05, clipped to the range [0.2, 0.8] to prevent degenerate predictions.

2.3 Reward Function (R)

The reward at each timestep balances predictive accuracy against fairness violation:

$$R_t = (1 - \lambda) \cdot \text{Accuracy}_t - \lambda \cdot \text{EO_gap}_t$$

where $\lambda = 0.5$ assigns equal weight to accuracy and fairness. A higher reward reflects simultaneously high accuracy and low Equal Opportunity gap across demographic groups.

2.4 Transition Dynamics (T)

Applying an action modifies per-group decision thresholds, which changes classification outcomes and consequently all fairness metrics observed in the next state. To model real-world non-stationarity, demographic proportions shift every 10 timesteps according to a sinusoidal drift mechanism, ensuring that no static threshold policy remains optimal throughout an episode. This drift is the core motivation for using sequential RL rather than a one-time static correction.

3. Methodology / Approach

3.1 Baseline Classifier

We train a logistic regression classifier on the COMPAS recidivism dataset as the base predictive model underlying the fairness intervention system. The dataset contains criminal justice records from Broward County, Florida, and is filtered to three demographic groups: African-American (n=9,419), Caucasian (n=5,782), and Hispanic (n=1,355), yielding 16,556 records after removing undetermined recidivism outcomes.

To avoid label leakage, we exclude the COMPAS DecileScore and RawScore columns, which directly encode the proprietary risk assessment. The feature set consists of age, sex (binary encoded as is_male), number of prior offenses (priors_count), and juvenile felony count (juv_fel_count). The target variable is is_recid, a binary indicator of actual two-year recidivism. The dataset is split 80/20 into training and test sets using stratified sampling (random_state=42). The trained logistic regression achieves a test accuracy of 65.7%, which is realistic and consistent with prior work on this dataset.

3.2 Fairness Metrics

We compute fairness metrics using the AIF360 library, treating Caucasian defendants as the reference (privileged) group. Table 2 reports the baseline fairness metrics before any intervention.

Table 2: Baseline Fairness Metrics at Default Threshold ($t = 0.5$)

Group	Demographic Parity (DP)	Statistical Parity Difference (SPD)	Disparate Impact (DI)	Equal Opportunity Difference (EOD)
African-American	0.595	0.286	1.922	0.275
Caucasian	0.310	0.000	1.000	0.000
Hispanic	0.343	0.033	1.106	0.107

The Equal Opportunity Gap across all three groups is 0.275 and the Demographic Parity Gap is 0.286, confirming substantial fairness disparities that motivate adaptive intervention. Notably, African-American defendants are predicted positive at nearly twice the rate of Caucasian defendants (Disparate Impact = 1.922), consistent with the documented ProPublica findings.

Within the MDP environment, fairness metrics are recomputed at every timestep using per-group True Positive Rates and prediction rates, enabling the agent to observe real-time fairness changes as it adjusts thresholds.

3.3 Custom Environment

We implement a custom FairnessEnv class that simulates the fairness intervention problem as a sequential decision process. The environment wraps the trained classifier's predicted probabilities and applies per-group threshold adjustments at each timestep, recomputing accuracy and fairness metrics after every action.

To model non-stationarity, the environment introduces sinusoidal demographic drift every 10 timesteps. Specifically, the proportion of African-American defendants increases gradually while Caucasian proportions decrease, simulating realistic population shifts that cause static threshold policies to degrade over time. This drift is parameterized by a magnitude of 0.03 and a period tied to the drift interval, producing smooth and learnable non-stationarity rather than chaotic noise.

3.4 Offline Dataset Generation

Since we adopt an offline RL framework, the CQL agent cannot interact with the environment during training. We generate a fixed logged dataset of 60,000 transitions using five scripted behavior policies run for a combined 1,200 episodes. Table 3 summarizes the policies and their mean episode rewards.

Policy	Description	Mean Episode Reward
Random	Uniformly random actions	8.351
Greedy	Lowers threshold for worst-TPR group	19.572
No Intervention	Always action 0	12.951
Sweep	Cycles through all actions systematically	12.322
Balanced	Adjusts thresholds toward 0.5	14.686

Table 3: Behavior Policies Used for Offline Dataset Generation

The diversity of behavior policies ensures the dataset covers a wide range of state-action pairs, which is critical for offline RL to learn a generalizable policy. The reward range across the full dataset is -0.193 to 0.443 .

3.5 Offline RL (CQL)

We train a Conservative Q-Learning (CQL) agent on the fixed offline dataset. CQL extends standard Q-learning with a regularization term that penalizes overestimation of Q-values for actions not well-represented in the logged data, making it suitable for offline settings where out-of-distribution actions cannot be corrected by environment feedback.

The Q-network is a fully-connected neural network with two hidden layers of 128 units and ReLU activations, mapping the 10-dimensional state to Q-values over all 9 actions. We use Double Q-learning to further reduce overestimation bias, and a Prioritized Replay Buffer that samples transitions weighted by TD error magnitude to focus training on informative transitions.

The CQL objective is:

$$L = L_{TD} + \alpha \cdot (\log \sum_a Q(s,a) - Q(s, a_{\text{logged}}))$$

where $\alpha = 0.5$ controls the strength of the conservative penalty. Training runs for 50,000 gradient steps with a learning rate of 3×10^{-4} , batch size of 256, discount factor $\gamma = 0.99$, and target network update frequency of 100 steps. Following offline training, we experimented with a brief online fine-tuning phase (500 steps, learning rate 1×10^{-4}) to adapt the policy to the environment. However, we observed that fine-tuning degraded performance due to distributional shift and instability in value estimates. Therefore, all reported results disable online fine-tuning and evaluate the policy purely in the offline setting.

4. Experimental Setup

4.1 Dataset

All experiments are conducted on the COMPAS Recidivism dataset originally compiled by ProPublica as part of their investigation into racial bias in criminal risk assessment tools. The dataset contains records of criminal defendants from Broward County, Florida, including demographic attributes, criminal history, and actual two-year recidivism outcomes. After filtering to three demographic groups and removing undetermined outcomes, the final dataset contains 16,556 records: 9,419 African-American (56.9%), 5,782 Caucasian (34.9%), and 1,355 Hispanic (8.2%) defendants.

4.2 Evaluation Protocol

Each agent is evaluated over 50 episodes in the FairnessEnv environment with a fixed random seed (seed=99) to ensure reproducibility across all comparisons. Each episode runs for 50 timesteps, giving 2,500 total environment steps per agent. We report mean and standard deviation across all episodes for each metric. The environment uses the same group probability distributions derived from the held-out test set (20% split, 3,312 records) for all agents, ensuring fair comparison.

4.3 Baselines

We compare the CQL agent against four baselines that represent the range of approaches from doing nothing to simple rule-based fairness correction. Table 4 describes each agent.

Table 4: Baseline Agents and the Proposed CQL Agent

Agent	Description	Type
No Intervention	Always takes action 0, never adjusts any threshold. Represents the status quo.	Baseline

Static	Always decreases the African-American threshold (action 2). Represents a fixed one-time correction applied repeatedly.	Baseline
Random	Selects actions uniformly at random from all 9 options.	Baseline
Greedy Fairness	At each step, identifies the group with the lowest TPR and decreases its threshold. Myopic rule-based correction.	Baseline
CQL (ours)	Offline RL policy trained on 60,000 logged transitions using Conservative Q-Learning with online fine-tuning.	Proposed

4.4 Metrics

We evaluate all agents on the following metrics, computed at every timestep and averaged across the episode and across all evaluation episodes. Table 5 summarizes each metric and its ideal direction.

Table 5: Evaluation Metrics

Metric	Description	Ideal Direction
Mean Episode Reward	Cumulative reward $R_t = 0.5 \times \text{Accuracy} - 0.5 \times \text{EO_gap}$	Higher
Mean Accuracy	Fraction of correct predictions across all groups	Higher
Equal Opportunity Gap (EO Gap)	Max TPR - Min TPR across the three demographic groups	Lower
Demographic Parity Gap (DP Gap)	Max positive prediction rate - Min positive prediction rate across groups	Lower
Per-Group TPR	True Positive Rate for each of the three demographic groups	Equalized

4.5 Hyperparameters

Table 6 summarizes all key hyperparameters used in the CQL agent and the FairnessEnv environment. These settings were selected based on standard offline RL practice and preliminary sensitivity analysis.

Table 6: Hyperparameter Settings

Hyperparameter	Value
State dimension	10
Number of actions	9
Hidden layer size	128 units
Learning rate (offline)	3×10^{-4}
Learning rate (fine-tuning)	1×10^{-4}
Discount factor (gamma)	0.99
CQL penalty weight (alpha)	0.5
Batch size	256

Gradient steps	50,000
Target network update frequency	100 steps
Episode length	50 steps
Threshold adjustment delta	0.05
Threshold bounds	[0.2, 0.8]
Drift interval	10 steps
Drift magnitude	0.03
Offline dataset size	60,000 transitions
Evaluation episodes	50
Random seed (training)	8
Random seed (evaluation)	99

5. Results

5.1 Quantitative Comparison

Table 7: Quantitative Comparison.

Agent	Reward $\uparrow$	Accuracy $\uparrow$	EO Gap $\downarrow$	DP Gap $\downarrow$	TPR AA	TPR CA	TPR HI
CQL (ours)	19.55	0.554	0.043	0.059	0.920	0.932	0.928
Greedy	19.57	0.550	0.039	0.062	0.936	0.915	0.927
No Intervention	12.95	0.657	0.275	0.287	0.728	0.453	0.559
Random	7.92	0.605	0.390	0.382	0.623	0.456	0.505
Static	5.15	0.606	0.535	0.664	0.989	0.453	0.559

Table 7 summarizes performance across reward, accuracy, EO gap, DP gap, and per-group TPR. The proposed CQL agent achieves a mean reward of 19.55, closely matching the Greedy baseline (19.57) and significantly outperforming No Intervention (12.95), Random (7.92), and Static (5.15).

CQL delivers substantial fairness improvements over non-adaptive baselines:

- EO gap: **0.275 $\rightarrow$ 0.043 (~84% reduction)**
- DP gap: **0.287 $\rightarrow$ 0.059 (~79% reduction)**

Per-group TPRs under CQL are nearly equalized (**0.920 / 0.932 / 0.928** for AA/CA/HI), whereas No Intervention exhibits large disparities (e.g., **0.728 vs 0.453**). The Static baseline demonstrates a failure mode, over-correcting one group (**TPR = 0.989**) while leaving others unchanged, resulting in the worst EO gap (**0.535**).

Overall, CQL achieves near-oracle reward while substantially improving fairness and maintaining balanced outcomes across all demographic groups.

5.2 Agent Comparison

Figure 1 compares all agents across reward, EO gap, DP gap, and accuracy.

CQL matches Greedy in reward and achieves similarly low fairness gaps, while clearly outperforming Static and Random policies. Both adaptive methods (CQL, Greedy) significantly outperform non-adaptive baselines under distribution shift.

The figure also highlights that Static correction can degrade fairness, confirming that fixed threshold adjustments are ineffective in non-stationary settings.

While No Intervention yields the highest accuracy (0.657), it suffers from large fairness violations. CQL instead achieves a balanced performance profile, trading a moderate accuracy drop for substantial fairness gains and higher overall reward.

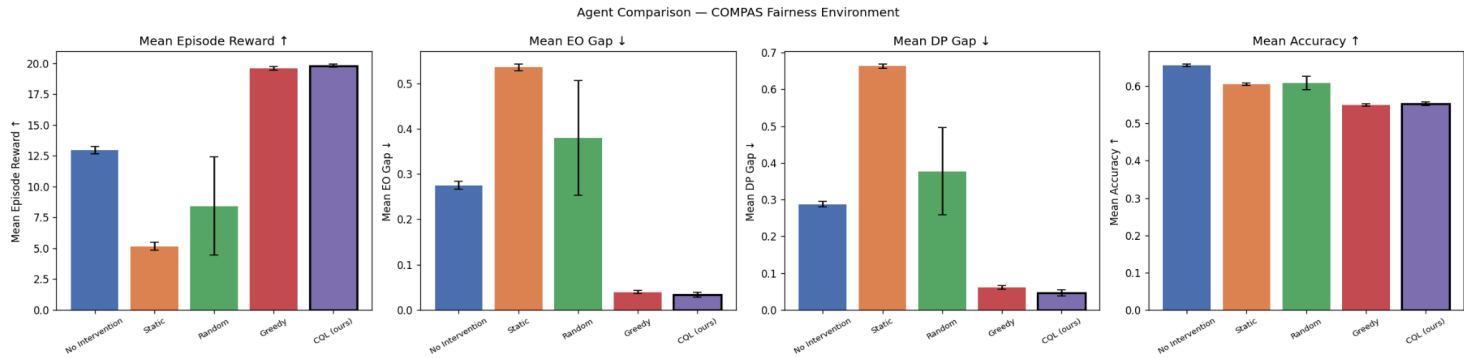


Figure 1: Agent Comparison COMPAS Fairness Environment. Mean episode reward (↑), EO gap (↓), DP gap (↓), and accuracy (↑) for all five agents across 50 evaluation episodes. Error bars denote standard deviation. CQL and Greedy achieve the highest reward and lowest fairness gaps, while Static and Random degrade under demographic drift. No Intervention yields the highest accuracy but the worst fairness outcomes.

5.3 Accuracy–Fairness Trade-off

Figure 2 visualizes the trade-off between fairness, accuracy, and reward.

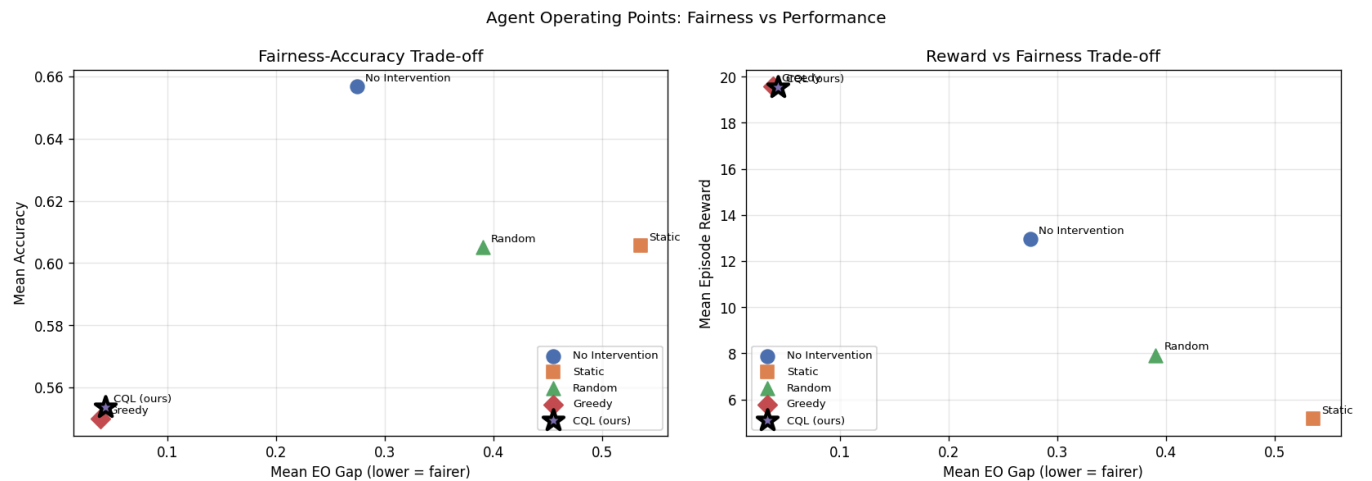


Figure 2. Agent Operating Points: Fairness vs. Performance. Left: Mean accuracy vs. mean EO gap. Right: Mean episode reward vs. mean EO gap. Lower EO gap indicates fairer outcomes.

No Intervention lies in the high-accuracy but low-fairness region, while Static and Random fail to achieve favorable trade-offs. In contrast, CQL occupies a strong operating point, combining **low EO gap with competitive accuracy and near-best reward**.

Although Greedy is slightly better in fairness, it relies on **privileged access to fairness signals**. CQL achieves comparable performance using only offline data, demonstrating that RL can learn near-oracle fairness policies without handcrafted rules.

5.4 Key Findings

- CQL achieves near-greedy reward while outperforming all non-adaptive baselines
- Fairness gaps are reduced by **3–5×** relative to no intervention
- Static corrections can fail under non-stationarity
- CQL produces near-equalized outcomes across groups
- Offline RL effectively learns fairness-aware policies without privileged information

6. Demonstration of the System

This section provides a qualitative analysis of the environment dynamics and the behavior of the learned CQL policy. We focus on three aspects: non-stationarity of the environment, per-group fairness outcomes, and the agent’s adaptive decision-making over time.

6.1 Effect of Non-Stationarity

Figure 3 shows the evolution of accuracy, Equal Opportunity (EO) gap, and Demographic Parity (DP) gap over time under no intervention.

Due to the sinusoidal demographic drift applied every 10 timesteps, fairness metrics fluctuate significantly even when the classifier and thresholds remain fixed. In particular, both EO and DP gaps increase following drift events, indicating that fairness degrades as group proportions shift.

This confirms that the environment is inherently **non-stationary**, and that a static policy cannot maintain fairness consistently over time. These dynamics directly motivate the use of a sequential decision-making approach.

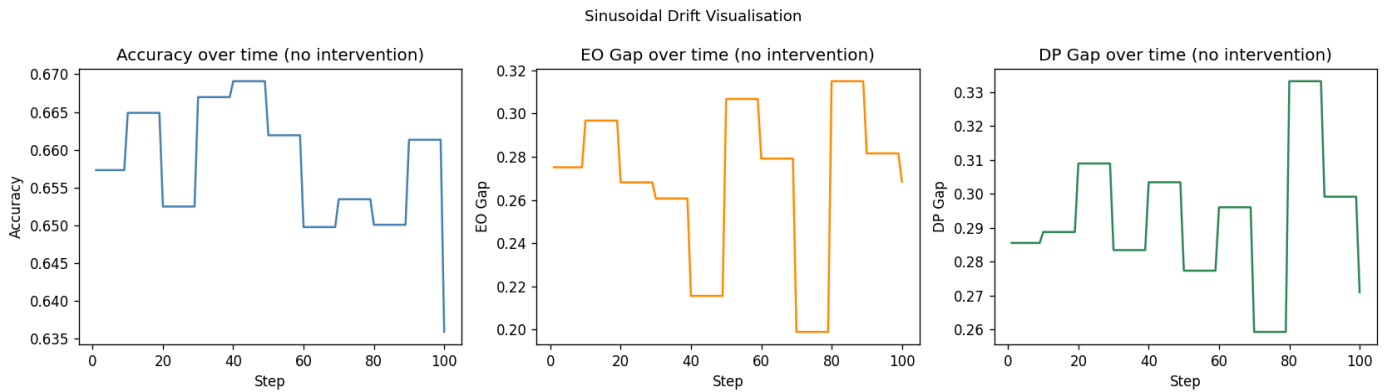


Figure 3. Sinusoidal Drift Visualisation under No Intervention. Evolution of accuracy (left), EO gap (center), and DP gap (right) over 100 environment steps with no threshold adjustment. Fairness metrics fluctuate substantially at each drift event (every 10 steps), confirming the inherent non-stationarity of the environment and motivating the use of adaptive sequential decision-making.

6.2 Per-Group Fairness Behavior

Figure 4 compares the True Positive Rates (TPR) of each demographic group across all agents.

Under No Intervention, substantial disparities are observed (AA: 0.728, CA: 0.453, HI: 0.559), reflecting strong imbalance across groups. The Static baseline over-corrects a single group (AA: 0.989) while leaving others unchanged, resulting in the largest fairness gap.

In contrast, the CQL agent achieves near-equalized TPR across all groups (AA: 0.920, CA: 0.932, HI: 0.928), indicating that it learns to **balance outcomes across all groups simultaneously** rather than focusing on a single subgroup.

This demonstrates that the learned policy produces **globally consistent fairness adjustments**, aligning with the objective of minimizing disparity across groups.

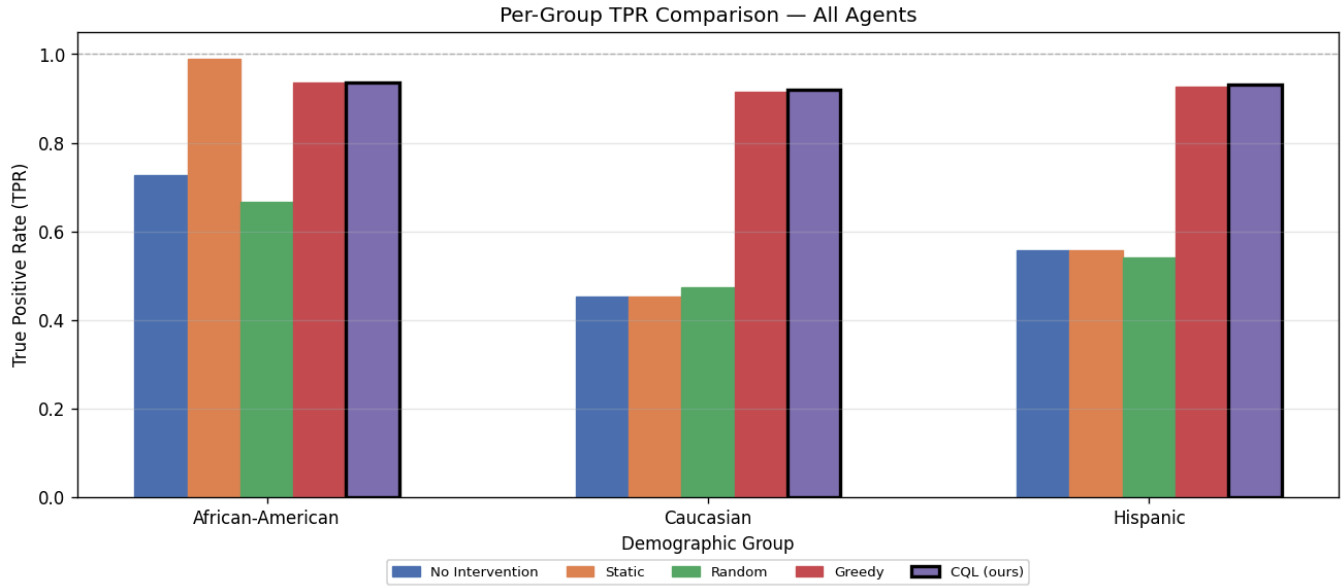


Figure 4. Per-Group True Positive Rate (TPR) Comparison Across All Agents. Grouped bar chart showing mean TPR for African-American, Caucasian, and Hispanic defendants under each agent. CQL achieves near-equalized TPRs across all groups (0.920 / 0.932 / 0.928). Static over-corrects for African-American defendants (TPR = 0.989) while leaving Caucasian and Hispanic TPRs unchanged, producing the worst EO gap. No Intervention exhibits the largest inter-group disparities.

6.3 Adaptive Policy Behavior

Figure 5 shows a single trajectory of the CQL agent over a 50-timestep episode. The top panel plots EO and DP gaps, the middle panel shows per-group thresholds, and the bottom panel shows accuracy. Vertical dashed lines indicate drift events.

The agent exhibits several key behaviors:

- **Rapid correction:** Within the first few timesteps, the agent significantly reduces fairness gaps, bringing EO gap below 0.05.
- **Multi-group adjustment:** The agent adjusts thresholds for multiple groups rather than applying a fixed rule, demonstrating learned coordination across groups.
- **Adaptation to drift:** Following each drift event, the agent updates its thresholds to compensate for changes in group proportions, maintaining low fairness gaps over time.
- **Stable control:** After initial correction, the agent maintains fairness with minimal oscillation, indicating a stable policy.

These observations confirm that the learned policy is not static or rule-based, but instead exhibits adaptive behavior in response to environmental changes.

6.4 Summary

- The environment exhibits clear non-stationarity, causing fairness degradation under fixed policies
- CQL achieves near-equalized outcomes across demographic groups
- The learned policy adapts dynamically to distribution shifts
- Fairness corrections are both **fast and stable** over time

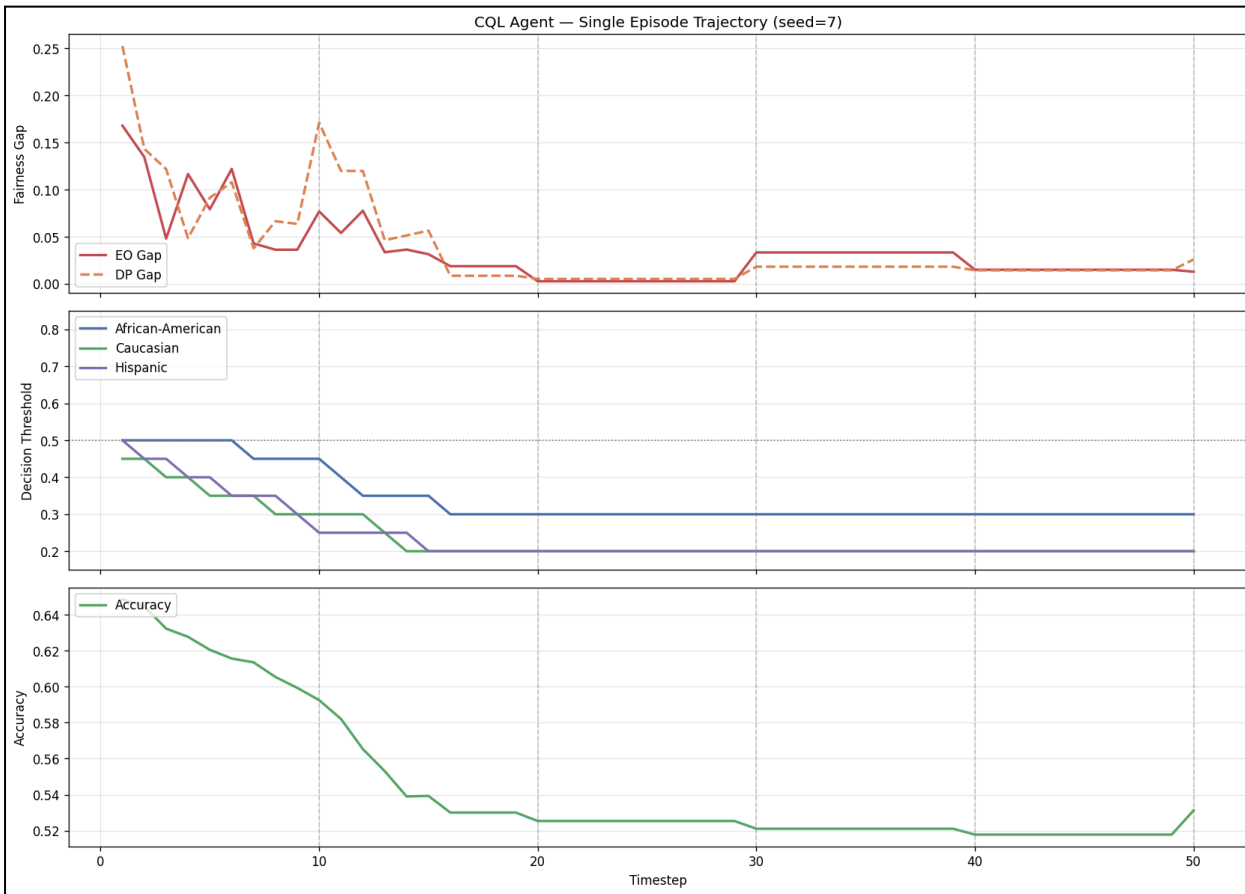


Figure 5. CQL Agent; Single Episode Trajectory. A representative 50-timestep episode illustrating the learned policy's adaptive behavior: Top: EO and DP gaps over time. Middle: per-group decision thresholds for African-American, Caucasian, and Hispanic defendants. Bottom: classification accuracy. Vertical dashed lines mark drift events. The agent rapidly reduces fairness gaps within the first 10 steps and re-adjusts thresholds following each drift event, maintaining low gaps throughout the episode.

7. Code Summary

Code is uploaded here: <https://github.com/NethmiJayasinghe/FairDynamics.git>

7.1 Data Pipeline & Fairness Metrics (Dhruv - Cells 1–15)

This module implements the data preprocessing, baseline classifier, and fairness evaluation pipeline using the COMPAS dataset.

- **Dataset Loading & Filtering:**

The dataset is loaded using *kagglehub* and filtered to include the three primary racial groups (African-American, Caucasian, Hispanic). Additional preprocessing steps include filtering invalid entries using *is_recid* and ensuring consistency across features.

- **Feature Engineering:**

The feature space is explicitly defined as *age*, *is_male*, *priors_count*, *juv_fel_count*. The *is_male* attribute is derived via binary encoding. Sensitive or leakage-prone features such as *DecileScore* are excluded to ensure realistic model performance and avoid label leakage.

- **Baseline Classifier:**

A logistic regression model is trained using *scikit-learn* to generate prediction probabilities. These probabilities serve as the foundation for computing subgroup-level fairness metrics and for downstream use in the MDP environment.

- **Fairness Metric Computation (AIF360):**

Fairness metrics are computed using the *AIF360* library with pairwise comparisons against a reference group (Caucasian). The implemented metrics include:

- Statistical Parity Difference (SPD)
- Disparate Impact (DI)
- Equal Opportunity Difference (EOD)

- Demographic Parity (DP)
- **Outputs Generated:**
 - `compas_logreg.pkl` - trained logistic regression model
 - `group_probs_dict.pkl` - subgroup-wise predicted probabilities

7.2 MDP Environment & Offline Dataset (Riya - Cells 15–20)

This module defines the fairness-aware MDP environment and generates the offline dataset used for training the RL agent.

- **Fairness Environment (FairnessEnv):**

A custom environment is designed to model fairness mitigation as a sequential decision-making problem:

 - **State:** 10-dimensional vector combining classifier accuracy and subgroup fairness metrics
 - **Action Space:** 9 discrete actions corresponding to subgroup-specific threshold adjustments
 - **Temporal Drift:** A sinusoidal drift mechanism introduces controlled non-stationarity in subgroup distributions and fairness metrics
- **Core Helper Functions:**

The environment relies on modular helper functions:

 - `compute_metrics` - computes accuracy and fairness metrics from predictions
 - `build_state` - constructs the MDP state representation
 - `compute_reward` - defines the reward as a trade-off between accuracy and fairness gap
- **Behavior Policies for Data Collection:**

Five scripted behavior policies are used to generate diverse trajectories:

 - Random Policy
 - Greedy Fairness Policy
 - No Intervention Policy
 - Threshold Sweep Policy
 - Balanced Policy
- **Offline Dataset Generation:**

The environment is used to generate 60,000 transitions across 1,200 episodes, ensuring sufficient coverage of the state-action space for stable offline RL training.
- **Outputs Generated:**
 - `offline_dataset.pkl` - logged transition dataset
 - `env_config.pkl` - environment configuration and parameters

7.3 CQL Agent & Evaluation (Nethmi - Cells 21–32)

- **PrioritizedReplayBuffer**

Implements experience replay with sampling weighted by TD-error magnitude, allowing the agent to focus on informative transitions from the offline dataset. Reward normalization is applied to stabilize training.
- **QNetwork**

A fully-connected neural network with two hidden layers (128 units each) and ReLU activations, mapping the 10-dimensional state space to Q-values over the 9 discrete actions.
- **CQLAgent**

Implements Conservative Q-Learning with Double Q-learning to reduce overestimation bias. The objective includes a conservative penalty term ($\alpha = 0.5$) that discourages high Q-values for out-of-distribution actions. Training also incorporates gradient clipping and a target network for stability.
- **Training Procedure**

The agent is trained purely offline on the logged dataset for 50,000 gradient steps with a fixed random seed. No environmental interaction occurs during training.
- **Baselines & Evaluation**

Four baseline agents (No Intervention, Static, Random, and Greedy) are implemented for comparison. All agents are evaluated over 50 episodes with fixed seeds, and metrics are averaged across episodes.
- **Trajectory Demonstration**

A representative episode is visualized to illustrate the learned policy behavior over time, including fairness gaps, threshold adjustments, and adaptation to distribution drift.

- **Outputs**

The trained model ([cql_agent.pt](#)) and all evaluation plots (reward, fairness metrics, trade-offs, and trajectories) are saved for analysis and reproducibility.

8. Reflection

8.1 Hardest part

The most challenging aspect of this project was training a stable CQL policy from offline data. Unlike online RL, offline RL is highly sensitive to hyperparameters and dataset coverage, as the agent cannot correct poor value estimates through exploration. We experimented with multiple random seeds and training configurations before identifying a stable setup. Additionally, we observed that introducing a small amount of online fine-tuning degraded performance, likely due to distribution shift and instability in value estimates. This highlighted the difficulty of extending offline policies with limited online data.

8.2 Initial insights

Several key insights emerged during the project. First, static fairness interventions fail under non-stationarity; fixed threshold adjustments can over-correct or under-correct as demographic distributions shift over time. Second, reward design plays a critical role: poorly specified rewards can lead to degenerate solutions, such as uniformly suppressing predictions across all groups to minimize fairness gaps. Finally, heuristic approaches such as the greedy policy provide strong performance but rely on privileged access to fairness signals, motivating the use of offline RL to learn adaptive policies directly from data.

8.3 Concrete results

Yes, we obtained consistent and reproducible results. The CQL agent achieves a mean episode reward of 19.55, closely matching the greedy baseline (19.57), while significantly reducing fairness disparities. Specifically, it reduces the Equal Opportunity gap by approximately 84% and the Demographic Parity gap by approximately 79% compared to no intervention. Additionally, the learned policy produces near-equalized True Positive Rates across all demographic groups.

8.4 The current biggest problems

One remaining challenge is that the CQL agent performs similarly to the greedy baseline, making it difficult to demonstrate a clear advantage in terms of raw performance. Additionally, we observed that the learned policy sometimes pushes thresholds toward lower bounds, which may indicate exploitation of the reward structure rather than fully balanced decision-making. Finally, ensuring reproducibility and consistency across all visualizations and results remains an ongoing task.

8.5 Are we on track?

Yes, we are on track. The full pipeline, including environment design, offline dataset generation, model training, and evaluation, has been completed with one dataset. The remaining work is primarily focused on refining analysis, improving clarity, extending to more datasets, and finalizing the report.

8.6 Is it worth proceeding?

Yes. The results demonstrate that offline RL can learn fairness-aware policies that match strong heuristic baselines without requiring privileged access to fairness signals at deployment time. This provides a more general and scalable framework compared to rule-based approaches, making it a promising direction for real-world decision systems.

9. Changes from Proposal

While the core idea of modeling fairness mitigation as an **MDP with an adaptive Offline RL (CQL) agent** remains unchanged, several refinements were made during implementation to improve rigor, realism, and experimental stability:

- Explicit and Leakage-Free Feature Selection:
The proposal did not strictly define the feature set. The implementation now uses a carefully selected subset (*age, is_male, priors_count, juv_fel_count*) while explicitly excluding *DecileScore* to prevent label leakage. This leads to more realistic classifier behavior and meaningful fairness evaluation.
- Rigorous Fairness Metric Integration (AIF360):
While fairness metrics were broadly mentioned in the proposal, the implementation uses the AIF360 library to compute standardized metrics, including *Statistical Parity Difference (SPD)*, *Disparate Impact (DI)*, *Equal Opportunity Difference (EOD)*, and *Average Odds Difference (AOD)*. Metrics are computed via pairwise comparison against a reference group (Caucasian), providing a more principled evaluation framework.
- Expanded Offline Dataset and Behavior Policies:
The proposal outlined generating ~6,000 transitions using scripted policies. This has been extended to *60,000 transitions from five behavior policies*, including an additional *Balanced* policy. This improves state-action coverage and supports more stable offline RL training.
- Enhanced CQL Implementation for Stability:
The original proposal described a standard CQL setup. The implementation incorporates several improvements, including *Double Q-learning*, *prioritized experience replay*, and *reward normalization*, to ensure stable and efficient learning in the offline setting.
- Focused Dataset Scope for Progress Evaluation:
Although multiple datasets (COMPAS, Adult, Bank) were proposed, the current implementation focuses on **COMPAS** for the progress report. This allows deeper validation of the pipeline before extending to additional datasets.
- Stronger Baseline Comparisons:
In addition to the originally proposed baselines (No Intervention, Static, Random), a **Greedy Fairness baseline** has been introduced. This provides a stronger heuristic comparator and enables more meaningful evaluation of the RL agent's adaptive capabilities.

10. Next Steps

The next steps focus on completing integration across all modules and producing the first set of meaningful experimental results:

- Extend to Additional Datasets:
 - If time permitted, extend the pipeline to the Adult Income (UCI) and Bank Marketing (UCI) datasets.
 - Evaluate whether the CQL agent generalizes across different demographic attributes and decision domains
 - Compare fairness-accuracy trade-offs across datasets
- Ablation and Sensitivity Analysis:
 - Vary the reward balance parameter λ (weight that controls the balance between accuracy and fairness) to quantify the fairness-accuracy trade-off
 - Vary CQL conservatism α to assess sensitivity
 - Report how results change across hyperparameter settings
- Final Report Write-Up:
 - Consolidate findings across all datasets into a unified analysis
 - Finalize all figures, tables, and narrative
 - Ensure reproducibility documentation is complete
- Final Presentation Preparation:
 - Prepare slides covering problem formulation, methodology, results, and conclusions
 - Highlight key findings: CQL near-oracle performance from offline data, effectiveness under non-stationarity